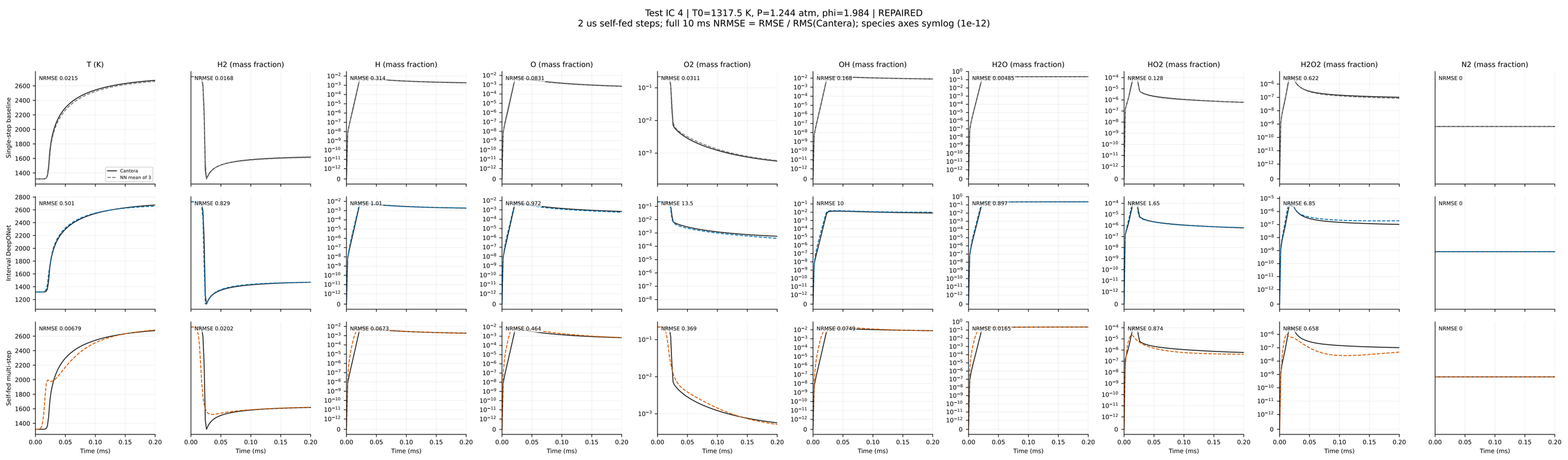
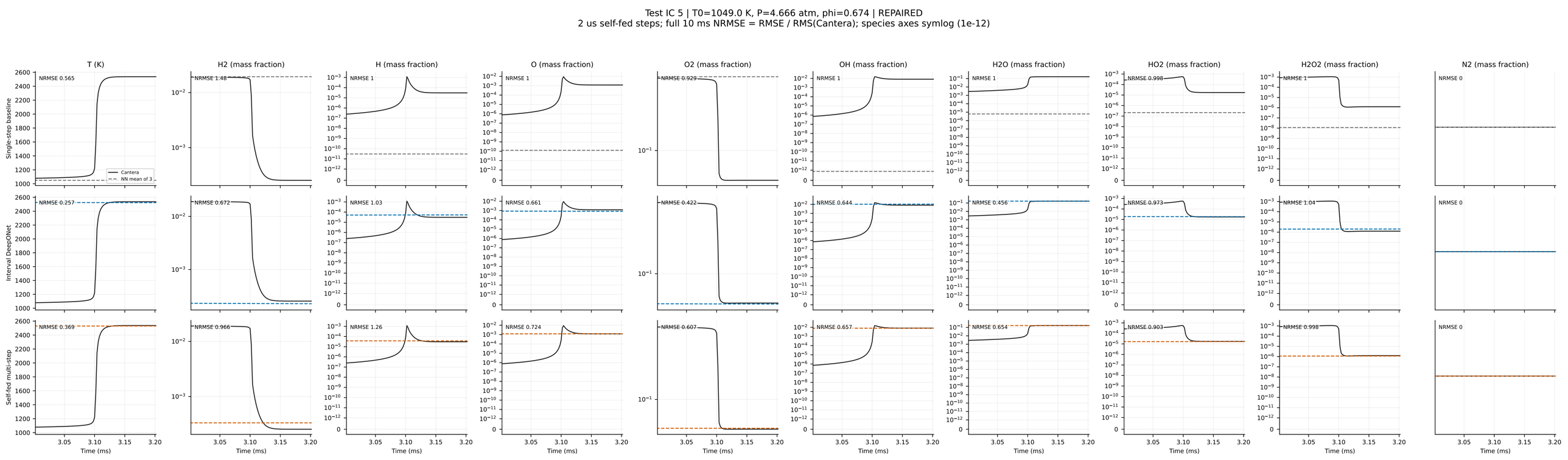
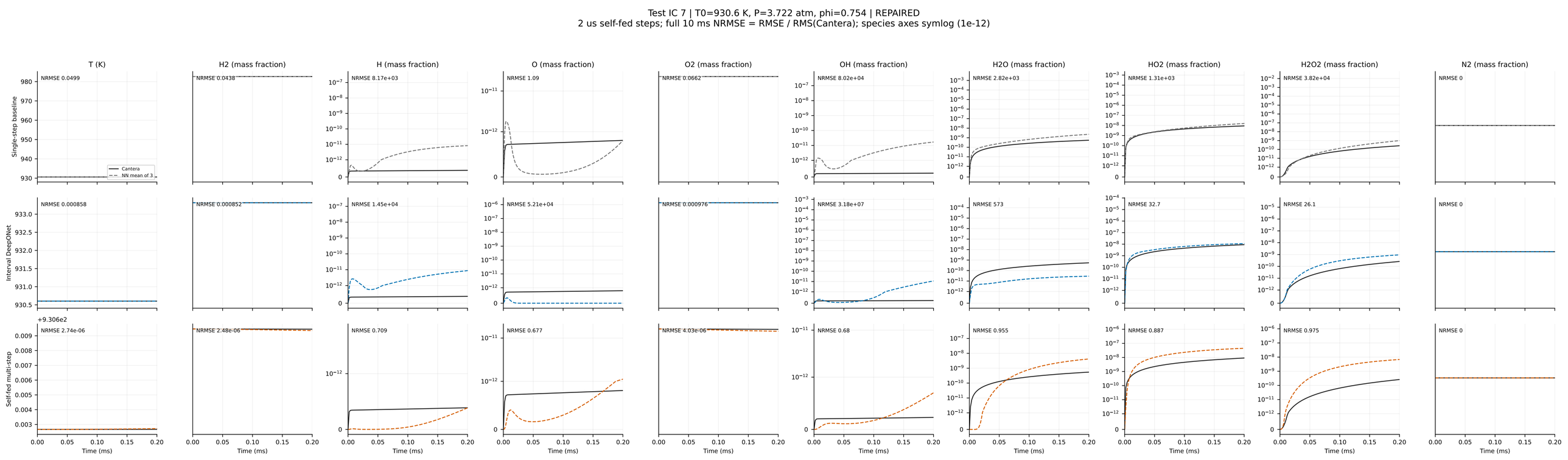
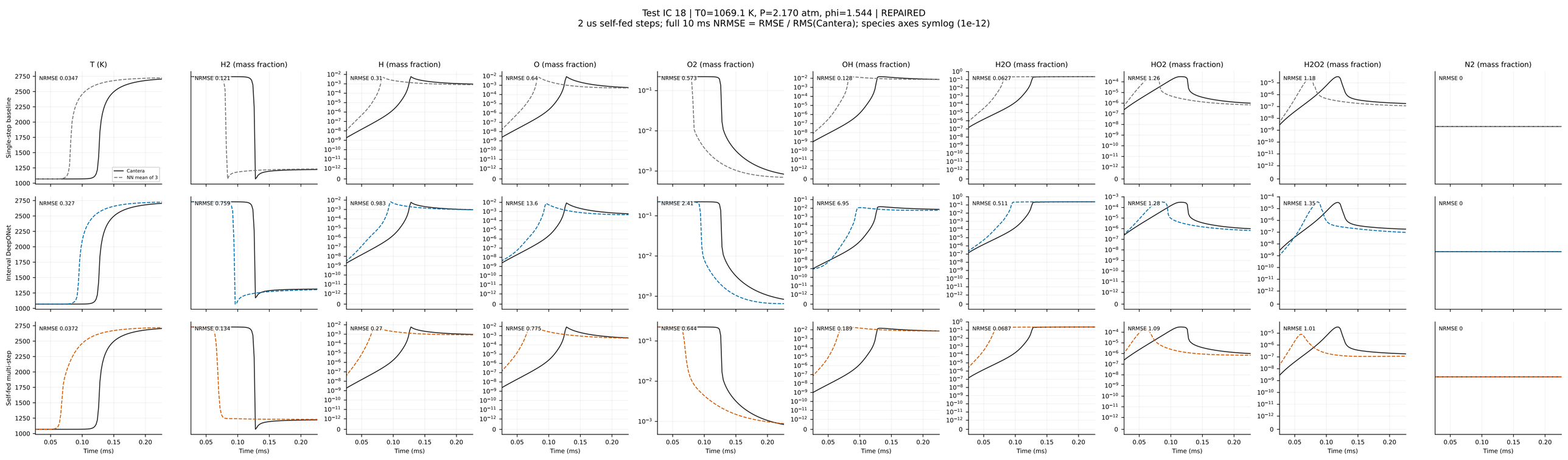
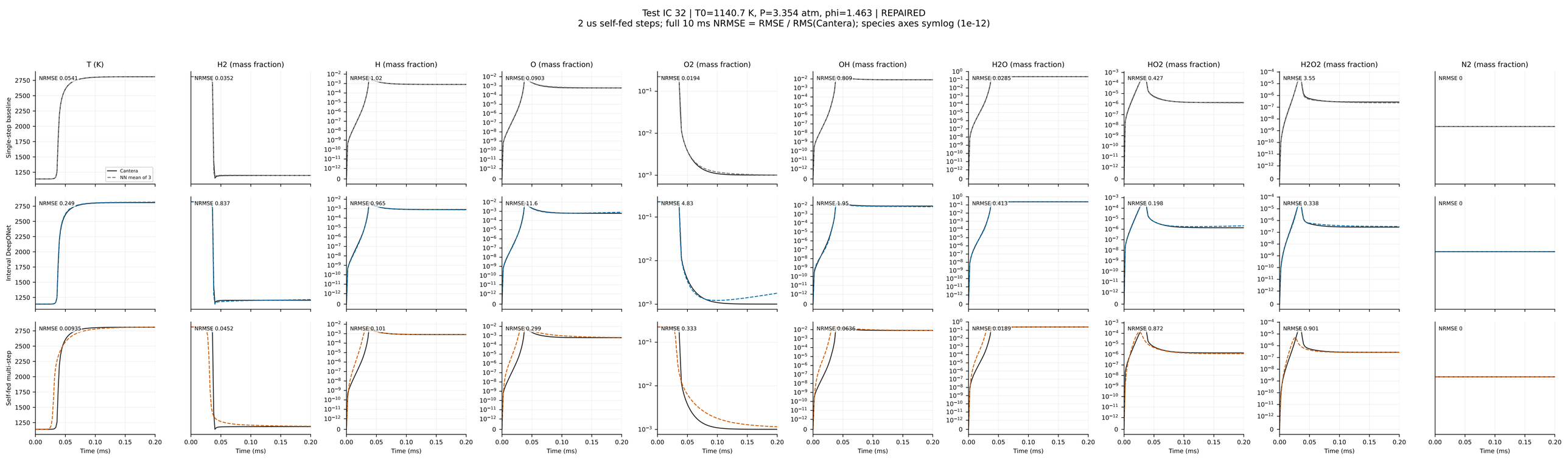
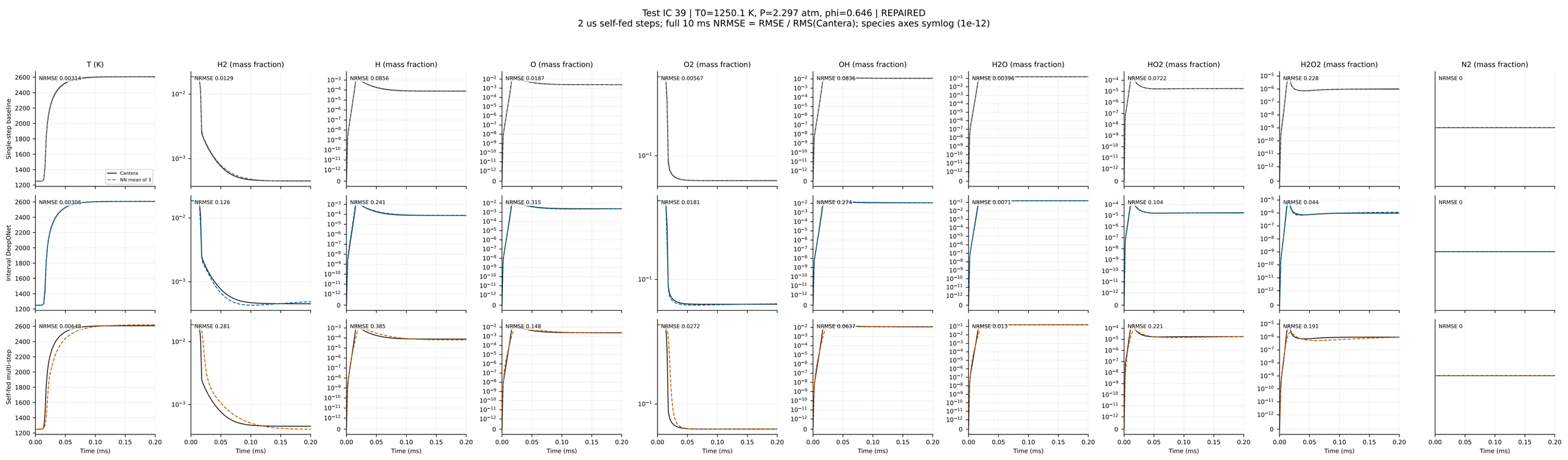

Test IC 4 | T0=1317.5 K, P=1.244 atm, phi=1.984 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0215
NRMSE 0.0168
NRMSE 0.314
NRMSE 0.0831
NRMSE 0.0311
NRMSE 0.168
NRMSE 0.00485
NRMSE 0.128
NRMSE 0.622
NRMSE 0
NRMSE 0.501
NRMSE 0.829
NRMSE 1.01
NRMSE 0.972
NRMSE 13.5
NRMSE 10
NRMSE 0.897
NRMSE 1.65
NRMSE 6.85
NRMSE 0
NRMSE 0.00679
NRMSE 0.0202
NRMSE 0.0673
NRMSE 0.464
NRMSE 0.369
NRMSE 0.0749
NRMSE 0.0165
NRMSE 0.874
NRMSE 0.658
NRMSE 0
Cantera
NN mean of 3
Time (ms)

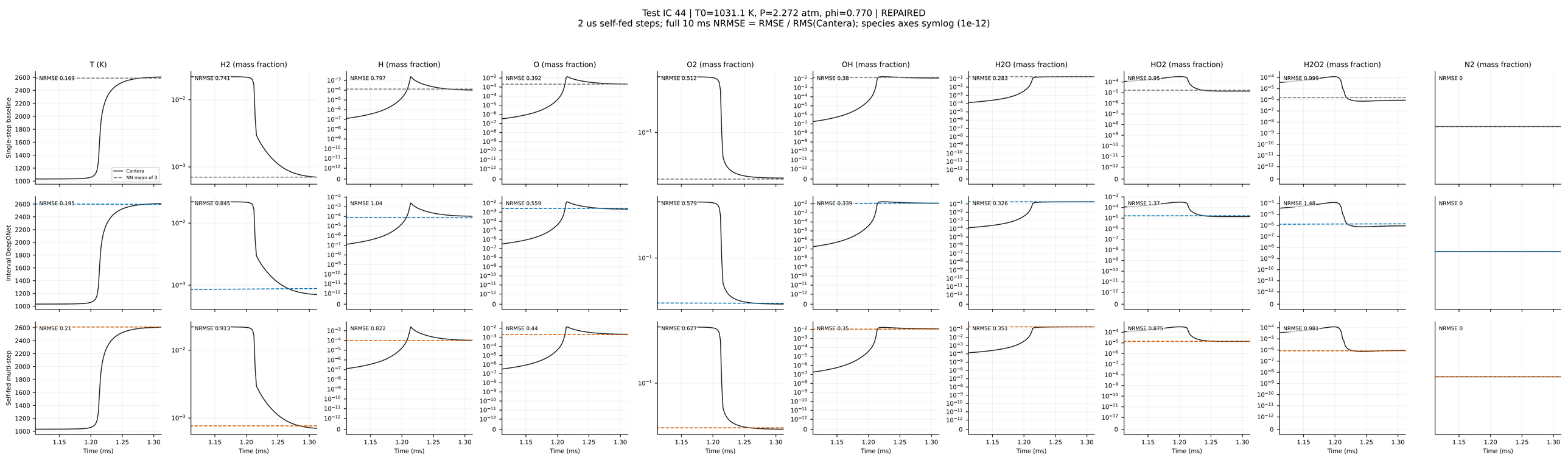

Test IC 5 | T0=1049.0 K, P=4.666 atm, phi=0.674 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.565
NRMSE 1.48
NRMSE 1
NRMSE 1
NRMSE 0.929
NRMSE 1
NRMSE 1
NRMSE 0.998
NRMSE 1
NRMSE 0
Cantera
NN mean of 3
NRMSE 0.257
NRMSE 0.672
NRMSE 1.03
NRMSE 0.661
NRMSE 0.422
NRMSE 0.644
NRMSE 0.456
NRMSE 0.973
NRMSE 1.04
NRMSE 0
NRMSE 0.369
NRMSE 0.966
NRMSE 1.26
NRMSE 0.724
NRMSE 0.607
NRMSE 0.657
NRMSE 0.654
NRMSE 0.903
NRMSE 0.998
NRMSE 0
Time (ms)

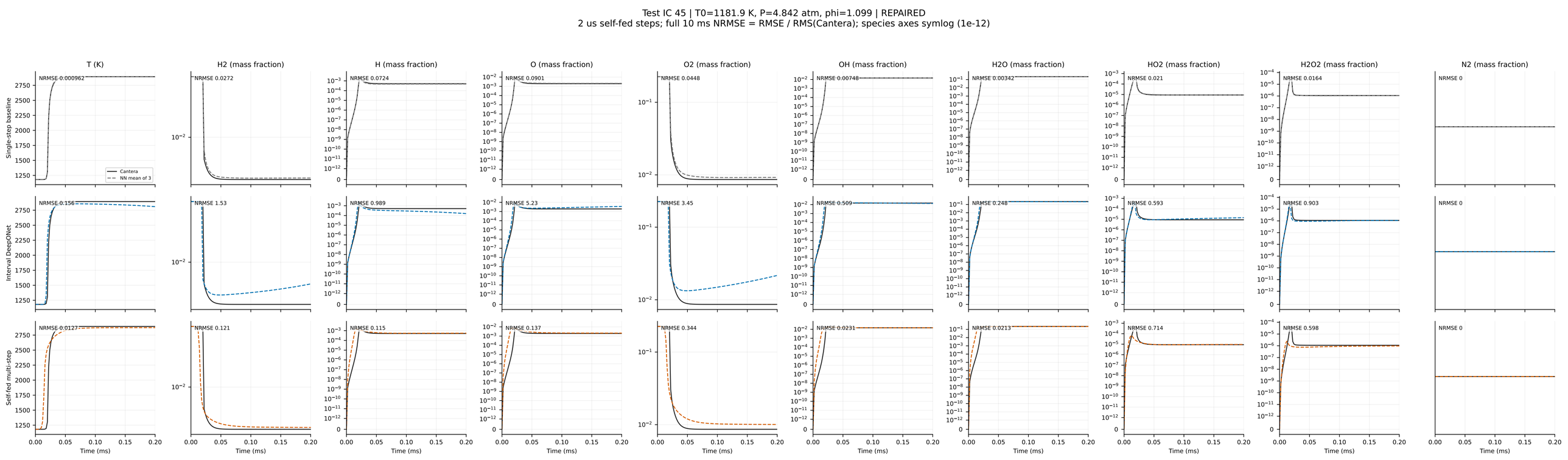

Test IC 7 | T0=930.6 K, P=3.722 atm, phi=0.754 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0499
NRMSE 0.0438
NRMSE 8.17e+03
NRMSE 1.09
NRMSE 0.0662
NRMSE 8.02e+04
NRMSE 2.82e+03
NRMSE 1.31e+03
NRMSE 3.82e+04
NRMSE 0
Cantera
NN mean of 3
NRMSE 0.000858
NRMSE 0.000852
NRMSE 1.45e+04
NRMSE 5.21e+04
NRMSE 0.000976
NRMSE 3.18e+07
NRMSE 573
NRMSE 32.7
NRMSE 26.1
NRMSE 0
+9.306e2
NRMSE 2.74e-06
NRMSE 2.48e-06
NRMSE 0.709
NRMSE 0.677
NRMSE 4.03e-06
NRMSE 0.68
NRMSE 0.955
NRMSE 0.887
NRMSE 0.975
NRMSE 0
Time (ms)

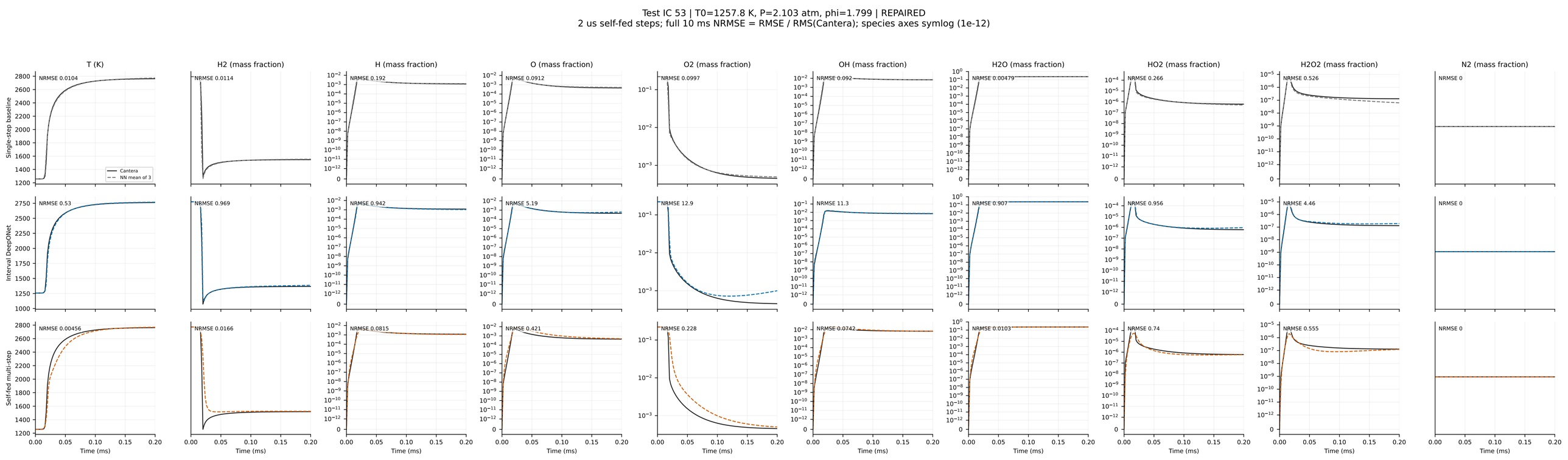

Test IC 18 | T0=1069.1 K, P=2.170 atm, phi=1.544 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0347
NRMSE 0.121
NRMSE 0.31
NRMSE 0.64
NRMSE 0.573
NRMSE 0.128
NRMSE 0.0627
NRMSE 1.26
NRMSE 1.18
NRMSE 0
Cantera
NN mean of 3
NRMSE 0.327
NRMSE 0.759
NRMSE 0.983
NRMSE 13.6
NRMSE 2.41
NRMSE 6.95
NRMSE 0.511
NRMSE 1.28
NRMSE 1.35
NRMSE 0
NRMSE 0.0372
NRMSE 0.134
NRMSE 0.27
NRMSE 0.775
NRMSE 0.644
NRMSE 0.189
NRMSE 0.0687
NRMSE 1.09
NRMSE 1.01
NRMSE 0
Time (ms)

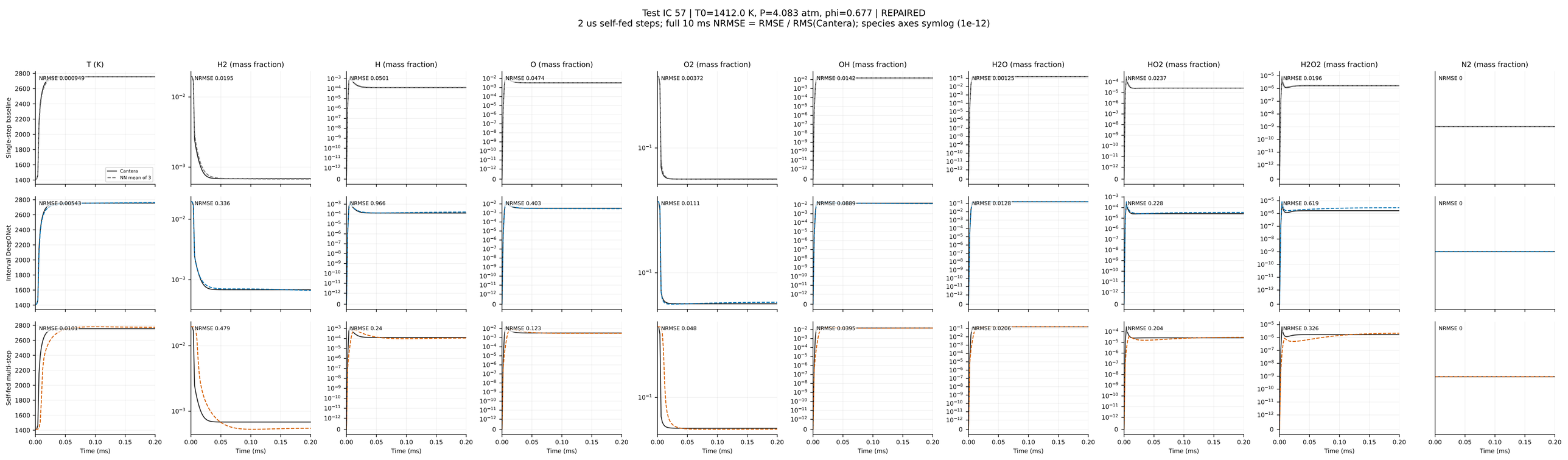

Test IC 32 | T0=1140.7 K, P=3.354 atm, phi=1.463 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0541
NRMSE 0.0352
NRMSE 1.02
NRMSE 0.0903
NRMSE 0.0194
NRMSE 0.809
NRMSE 0.0285
NRMSE 0.427
NRMSE 3.55
NRMSE 0
NRMSE 0.249
NRMSE 0.837
NRMSE 0.965
NRMSE 11.6
NRMSE 4.83
NRMSE 1.95
NRMSE 0.413
NRMSE 0.198
NRMSE 0.338
NRMSE 0
NRMSE 0.00935
NRMSE 0.0452
NRMSE 0.101
NRMSE 0.299
NRMSE 0.333
NRMSE 0.0636
NRMSE 0.0189
NRMSE 0.872
NRMSE 0.901
NRMSE 0
Cantera
NN mean of 3
Time (ms)

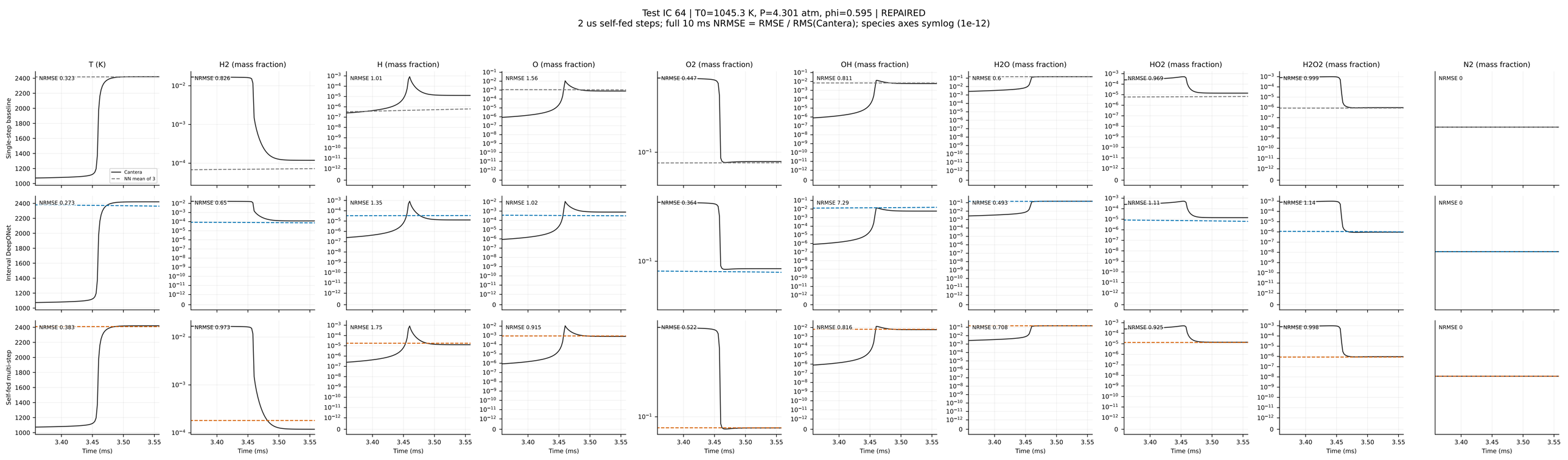

Test IC 39 | T0=1250.1 K, P=2.297 atm, phi=0.646 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.00314
NRMSE 0.0129
NRMSE 0.0856
NRMSE 0.0187
NRMSE 0.00567
NRMSE 0.0836
NRMSE 0.00396
NRMSE 0.0722
NRMSE 0.228
NRMSE 0
NRMSE 0.00306
NRMSE 0.126
NRMSE 0.241
NRMSE 0.315
NRMSE 0.0181
NRMSE 0.274
NRMSE 0.0071
NRMSE 0.104
NRMSE 0.044
NRMSE 0
NRMSE 0.00648
NRMSE 0.281
NRMSE 0.385
NRMSE 0.148
NRMSE 0.0272
NRMSE 0.0637
NRMSE 0.013
NRMSE 0.221
NRMSE 0.191
NRMSE 0
Time (ms)

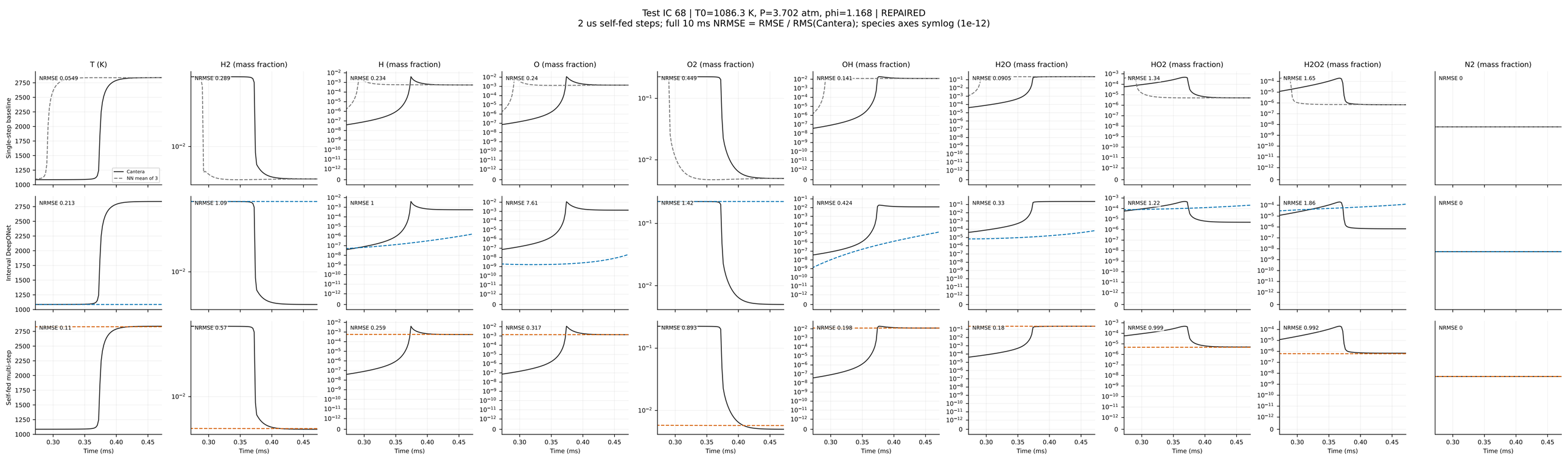

Test IC 44 | T0=1031.1 K, P=2.272 atm, phi=0.770 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.169
NRMSE 0.741
NRMSE 0.797
NRMSE 0.392
NRMSE 0.512
NRMSE 0.38
NRMSE 0.283
NRMSE 0.95
NRMSE 0.999
NRMSE 0
NRMSE 0.195
NRMSE 0.845
NRMSE 1.04
NRMSE 0.559
NRMSE 0.579
NRMSE 0.339
NRMSE 0.326
NRMSE 1.37
NRMSE 1.48
NRMSE 0
NRMSE 0.21
NRMSE 0.913
NRMSE 0.822
NRMSE 0.44
NRMSE 0.627
NRMSE 0.35
NRMSE 0.351
NRMSE 0.875
NRMSE 0.981
NRMSE 0
Cantera
NN mean of 3
Time (ms)

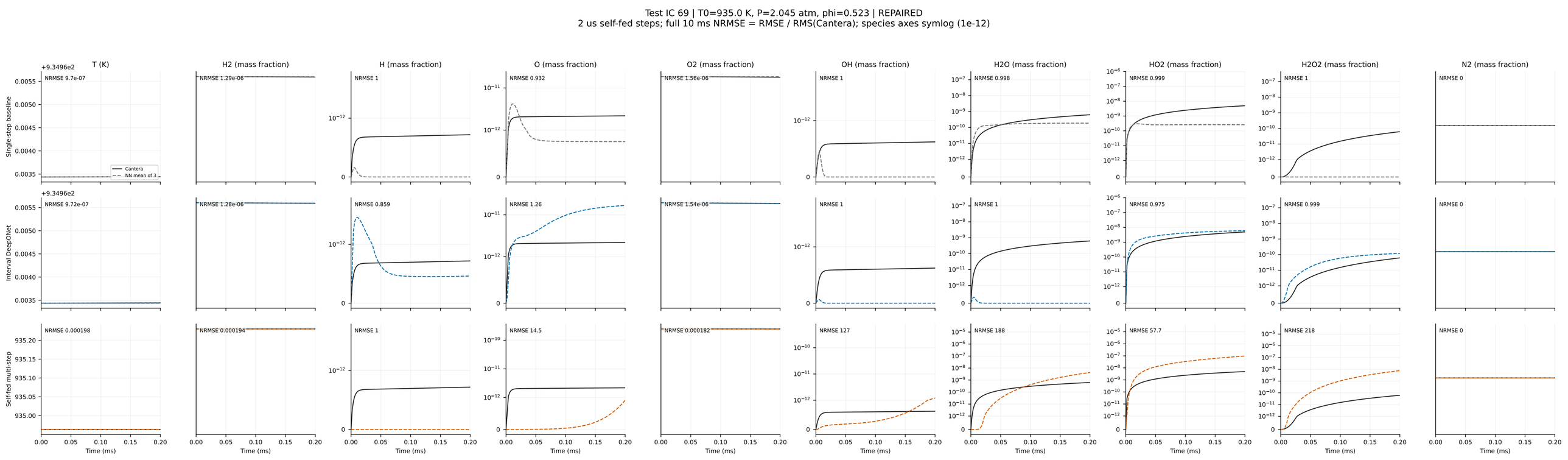

Test IC 45 | T0=1181.9 K, P=4.842 atm, phi=1.099 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.000962
NRMSE 0.0272
NRMSE 0.0724
NRMSE 0.0901
NRMSE 0.0448
NRMSE 0.00748
NRMSE 0.00342
NRMSE 0.021
NRMSE 0.0164
NRMSE 0
NRMSE 0.156
NRMSE 1.53
NRMSE 0.989
NRMSE 5.23
NRMSE 3.45
NRMSE 0.509
NRMSE 0.248
NRMSE 0.593
NRMSE 0.903
NRMSE 0
NRMSE 0.0127
NRMSE 0.121
NRMSE 0.115
NRMSE 0.137
NRMSE 0.344
NRMSE 0.0231
NRMSE 0.0213
NRMSE 0.714
NRMSE 0.598
NRMSE 0
Time (ms)

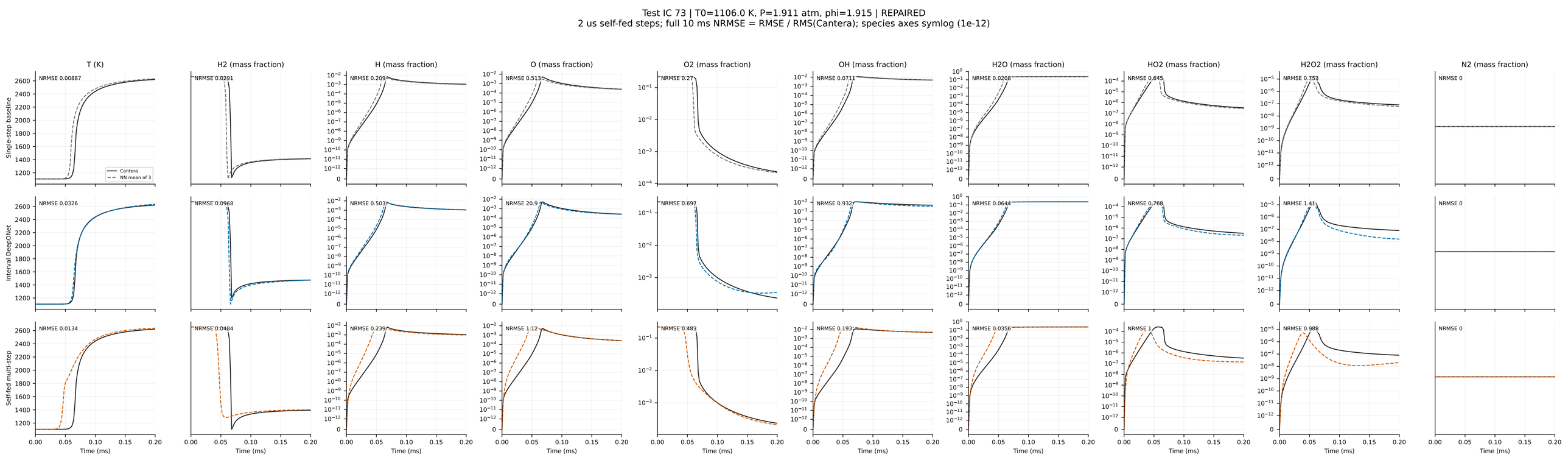

Test IC 53 | T0=1257.8 K, P=2.103 atm, phi=1.799 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0104
NRMSE 0.0114
NRMSE 0.192
NRMSE 0.0912
NRMSE 0.0997
NRMSE 0.092
NRMSE 0.00479
NRMSE 0.266
NRMSE 0.526
NRMSE 0
NRMSE 0.53
NRMSE 0.969
NRMSE 0.942
NRMSE 5.19
NRMSE 12.9
NRMSE 11.3
NRMSE 0.907
NRMSE 0.956
NRMSE 4.46
NRMSE 0
NRMSE 0.00456
NRMSE 0.0166
NRMSE 0.0815
NRMSE 0.421
NRMSE 0.228
NRMSE 0.0742
NRMSE 0.0103
NRMSE 0.74
NRMSE 0.555
NRMSE 0
Cantera
NN mean of 3
Time (ms)

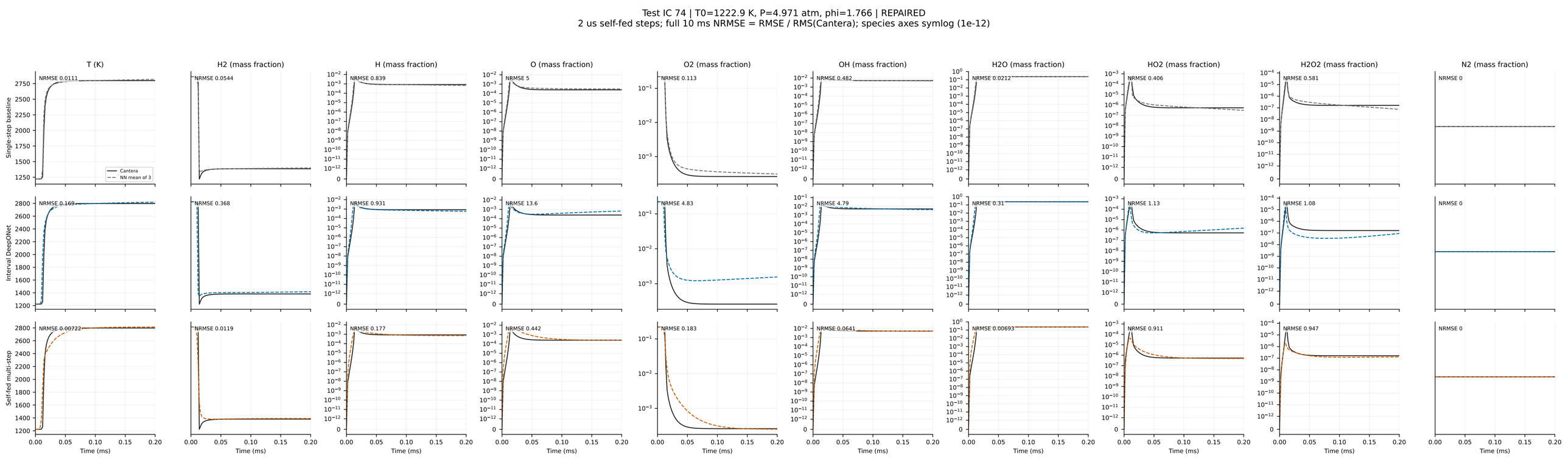

Test IC 57 | T0=1412.0 K, P=4.083 atm, phi=0.677 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.000949
NRMSE 0.0195
NRMSE 0.0501
NRMSE 0.0474
NRMSE 0.00372
NRMSE 0.0142
NRMSE 0.00125
NRMSE 0.0237
NRMSE 0.0196
NRMSE 0
NRMSE 0.00543
NRMSE 0.336
NRMSE 0.966
NRMSE 0.403
NRMSE 0.0111
NRMSE 0.0889
NRMSE 0.0128
NRMSE 0.228
NRMSE 0.619
NRMSE 0
NRMSE 0.0101
NRMSE 0.479
NRMSE 0.24
NRMSE 0.123
NRMSE 0.048
NRMSE 0.0395
NRMSE 0.0206
NRMSE 0.204
NRMSE 0.326
NRMSE 0
Time (ms)

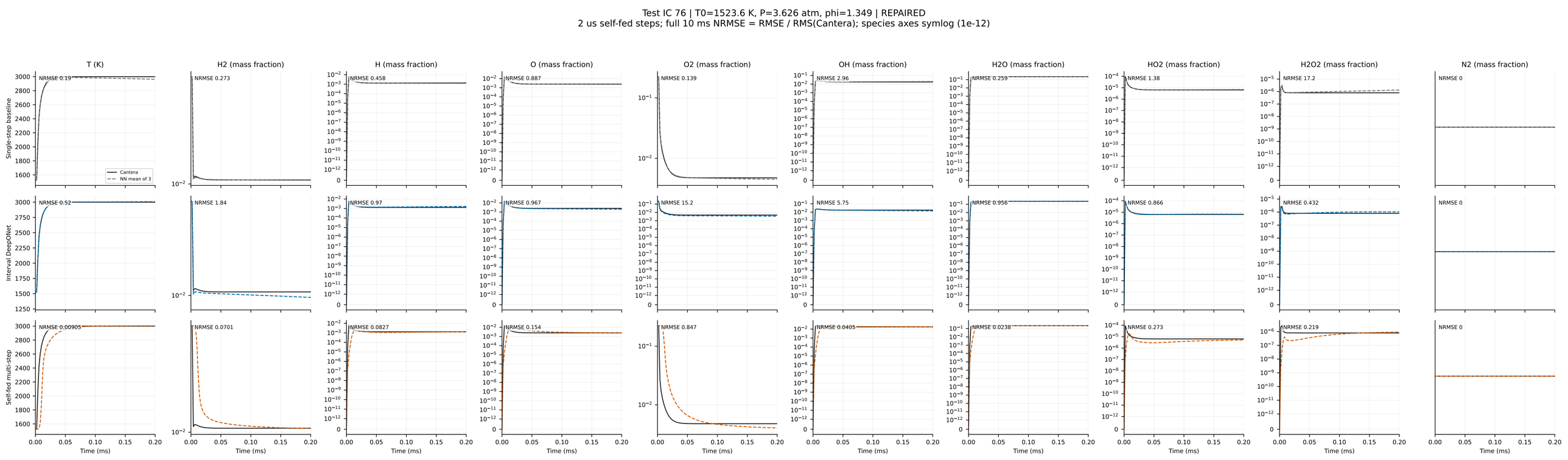

Test IC 64 | T0=1045.3 K, P=4.301 atm, phi=0.595 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.323
NRMSE 0.826
NRMSE 1.01
NRMSE 1.56
NRMSE 0.447
NRMSE 0.811
NRMSE 0.6
NRMSE 0.969
NRMSE 0.999
NRMSE 0
NRMSE 0.273
NRMSE 0.65
NRMSE 1.35
NRMSE 1.02
NRMSE 0.364
NRMSE 7.29
NRMSE 0.493
NRMSE 1.11
NRMSE 1.14
NRMSE 0
NRMSE 0.383
NRMSE 0.973
NRMSE 1.75
NRMSE 0.915
NRMSE 0.522
NRMSE 0.816
NRMSE 0.708
NRMSE 0.925
NRMSE 0.998
NRMSE 0
Cantera
NN mean of 3
Time (ms)

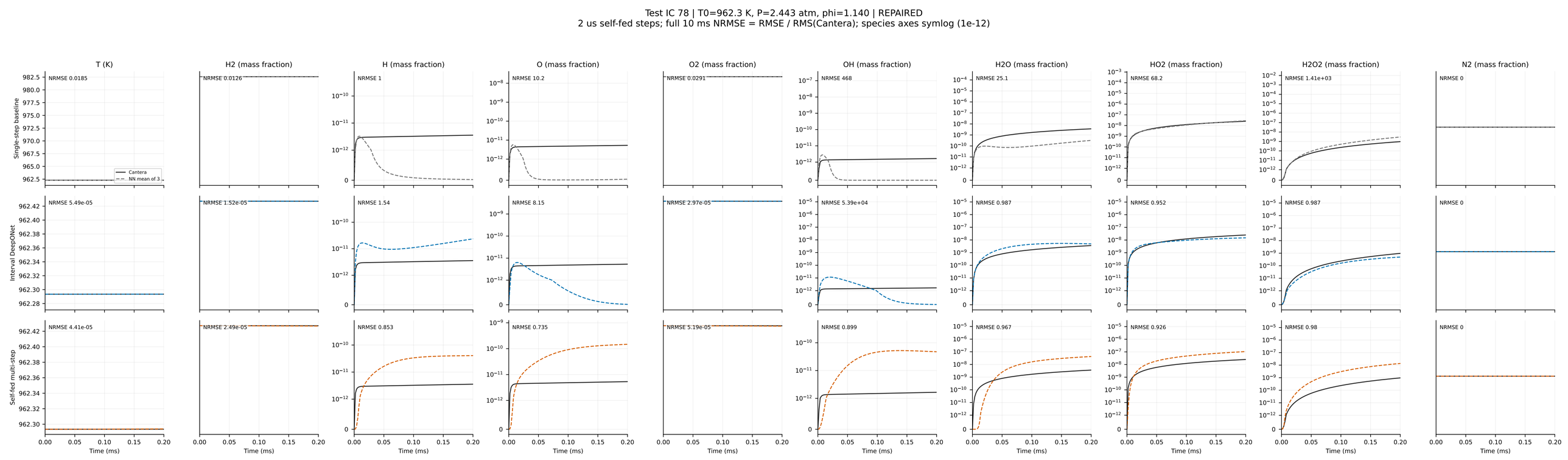

Test IC 68 | T0=1086.3 K, P=3.702 atm, phi=1.168 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0549
NRMSE 0.289
NRMSE 0.234
NRMSE 0.24
NRMSE 0.449
NRMSE 0.141
NRMSE 0.0905
NRMSE 1.34
NRMSE 1.65
NRMSE 0
Cantera
NN mean of 3
NRMSE 0.213
NRMSE 1.09
NRMSE 1
NRMSE 7.61
NRMSE 1.42
NRMSE 0.424
NRMSE 0.33
NRMSE 1.22
NRMSE 1.86
NRMSE 0
NRMSE 0.11
NRMSE 0.57
NRMSE 0.259
NRMSE 0.317
NRMSE 0.893
NRMSE 0.198
NRMSE 0.18
NRMSE 0.999
NRMSE 0.992
NRMSE 0
Time (ms)

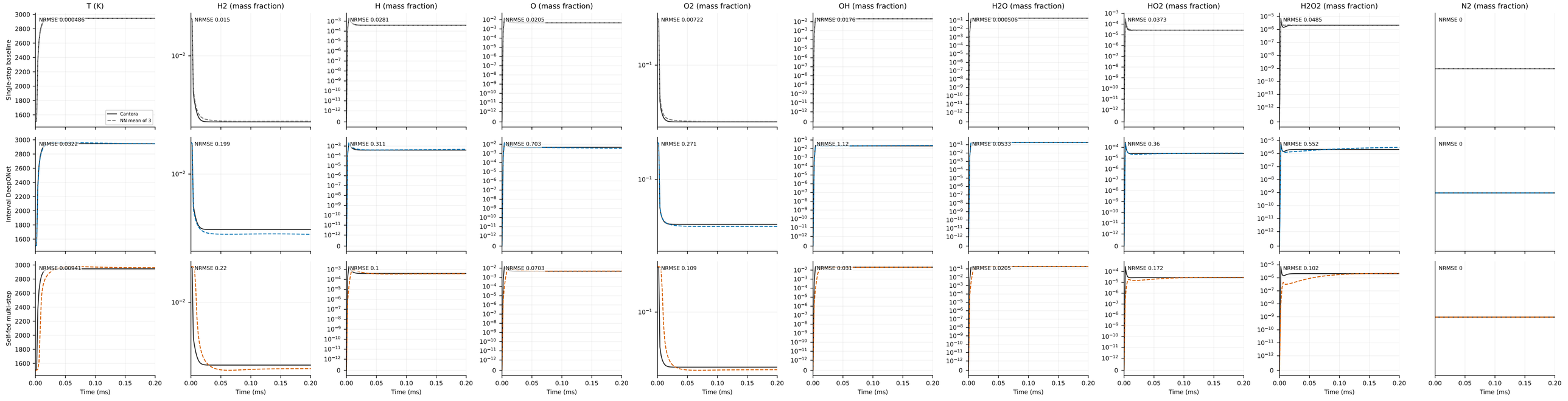

Test IC 69 | T0=935.0 K, P=2.045 atm, phi=0.523 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
Time (ms)


Test IC 73 | T0=1106.0 K, P=1.911 atm, phi=1.915 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.00887
NRMSE 0.0291
NRMSE 0.209
NRMSE 0.513
NRMSE 0.27
NRMSE 0.0711
NRMSE 0.0208
NRMSE 0.645
NRMSE 0.753
NRMSE 0
NRMSE 0.0326
NRMSE 0.0968
NRMSE 0.503
NRMSE 20.9
NRMSE 0.697
NRMSE 0.932
NRMSE 0.0644
NRMSE 0.768
NRMSE 1.41
NRMSE 0
NRMSE 0.0134
NRMSE 0.0484
NRMSE 0.239
NRMSE 1.12
NRMSE 0.483
NRMSE 0.193
NRMSE 0.0356
NRMSE 1
NRMSE 0.988
NRMSE 0
Cantera
NN mean of 3
Time (ms)


Test IC 74 | T0=1222.9 K, P=4.971 atm, phi=1.766 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0111
NRMSE 0.0544
NRMSE 0.839
NRMSE 5
NRMSE 0.113
NRMSE 0.482
NRMSE 0.0212
NRMSE 0.406
NRMSE 0.581
NRMSE 0
Cantera
NN mean of 3
NRMSE 0.169
NRMSE 0.368
NRMSE 0.931
NRMSE 13.6
NRMSE 4.83
NRMSE 4.79
NRMSE 0.31
NRMSE 1.13
NRMSE 1.08
NRMSE 0
NRMSE 0.00722
NRMSE 0.0119
NRMSE 0.177
NRMSE 0.442
NRMSE 0.183
NRMSE 0.0641
NRMSE 0.00693
NRMSE 0.911
NRMSE 0.947
NRMSE 0
Time (ms)


Test IC 76 | T0=1523.6 K, P=3.626 atm, phi=1.349 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.19
NRMSE 0.273
NRMSE 0.458
NRMSE 0.887
NRMSE 0.139
NRMSE 2.96
NRMSE 0.259
NRMSE 1.38
NRMSE 17.2
NRMSE 0
NRMSE 0.52
NRMSE 1.84
NRMSE 0.97
NRMSE 0.967
NRMSE 15.2
NRMSE 5.75
NRMSE 0.956
NRMSE 0.866
NRMSE 0.432
NRMSE 0
NRMSE 0.00905
NRMSE 0.0701
NRMSE 0.0827
NRMSE 0.154
NRMSE 0.847
NRMSE 0.0405
NRMSE 0.0238
NRMSE 0.273
NRMSE 0.219
NRMSE 0
Time (ms)


Test IC 78 | T0=962.3 K, P=2.443 atm, phi=1.140 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.0185
NRMSE 0.0126
NRMSE 1
NRMSE 10.2
NRMSE 0.0291
NRMSE 468
NRMSE 25.1
NRMSE 68.2
NRMSE 1.41e+03
NRMSE 0
NRMSE 5.49e-05
NRMSE 1.52e-05
NRMSE 1.54
NRMSE 8.15
NRMSE 2.97e-05
NRMSE 5.39e+04
NRMSE 0.987
NRMSE 0.952
NRMSE 0.987
NRMSE 0
NRMSE 4.41e-05
NRMSE 2.49e-05
NRMSE 0.853
NRMSE 0.735
NRMSE 5.19e-05
NRMSE 0.899
NRMSE 0.967
NRMSE 0.926
NRMSE 0.98
NRMSE 0
Time (ms)


Test IC 88 | T0=1502.1 K, P=4.980 atm, phi=0.859 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.000486
NRMSE 0.015
NRMSE 0.0281
NRMSE 0.0205
NRMSE 0.00722
NRMSE 0.0176
NRMSE 0.000506
NRMSE 0.0373
NRMSE 0.0485
NRMSE 0
NRMSE 0.0322
NRMSE 0.199
NRMSE 0.311
NRMSE 0.703
NRMSE 0.271
NRMSE 1.12
NRMSE 0.0533
NRMSE 0.36
NRMSE 0.552
NRMSE 0
NRMSE 0.00941
NRMSE 0.22
NRMSE 0.1
NRMSE 0.0703
NRMSE 0.109
NRMSE 0.031
NRMSE 0.0205
NRMSE 0.172
NRMSE 0.102
NRMSE 0
Time (ms)